

# ESET 355 LAB 6 REPORT

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|------------------|-------------------------------------------------------|
| Student name:    | Alex Armendariz<br>Easton Cavazos<br>Nikita Choudhary |
| Lab Section      | 502                                                   |
| Lab Name         | Lab 8: Antennas                                       |
| Lab session date | 12/2/2024                                             |
| Lab instructor   | Grant Gannon                                          |

## Discussion of Theory

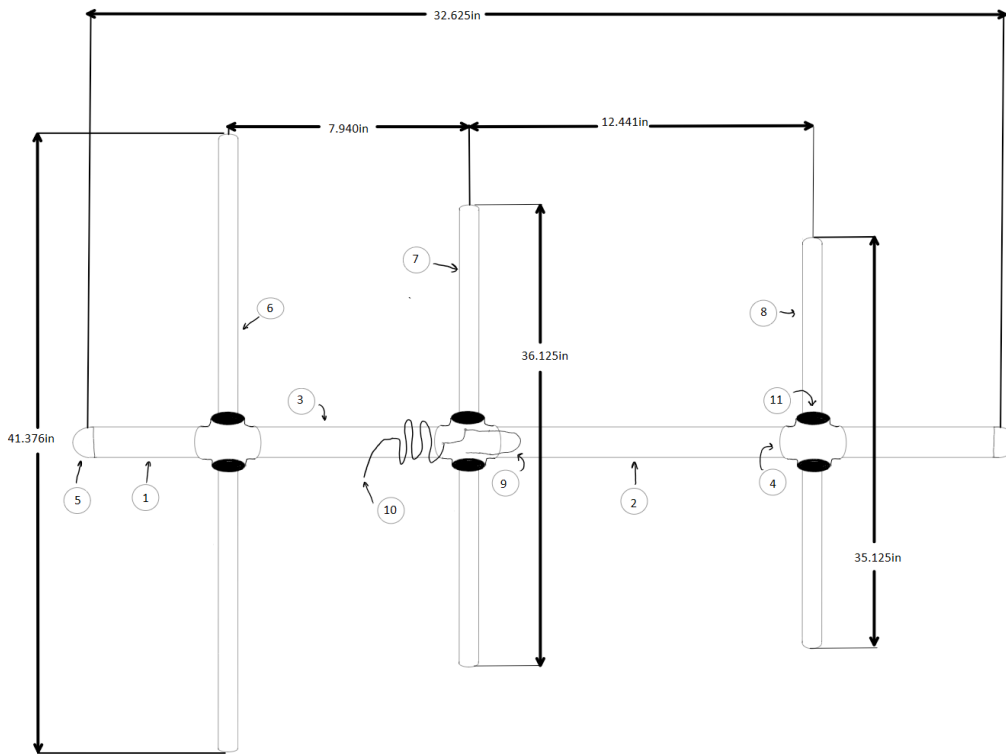
The type of antenna being used is a Yagi Antenna (a.k.a Yagi-Uda array, Beam antenna). These directional antennas are created using parallel rods that are connected and can usually be seen on top of large buildings. Although the antenna uses multiple rods only one is used as a driven element and receives a voltage/current while the other rods are used for reflection & direction. The length of the single reflector is made to be longer than the driven element to allow better reflection as well as make the impedance inductive. The antenna to the other side of the driver element rod is the direction rod. The direction rod is made to be shorter compared to the driver element to allow a capacitive inductance. Together the two rods will allow a phase distribution creating a traveling wave antenna that radiates across an axis (Bevelacqua, 2009).

## Construction and Drawings

While making the Yagi antenna, our frequency was kept in mind. Our antenna's frequency was chosen to be 146 MHz so that not only could it cooperate with a walkie-talkie, but so that the physical dimensions of the antenna itself weren't insanely massive. The Yagi antenna is made up of 3 pieces, or wires; the first wire is the reflector, the second wire is the dipole-driven element, and the third wire is the director.

As shown in Figure 1, the physical dimensions of the Yagi antenna are revealed. The 146 MHz frequency was used to calculate the wavelength by dividing the speed of light by this frequency to get a value of 2.0548 m. The wavelength was then used to calculate both the lengths of each wire and the space between them. Each of the 3 wires was made by cutting measuring tapes to their calculated lengths. The length of the reflector was calculated to be 41.376 inches, the length of the driven element to be 36.125 inches, and the length of the director to be 45.125 inches. However, because the driven element is separated, it is split in half into two equal pieces that are 17.750 inches each. These wires are all connected in parallel with the y-axis through  $\frac{1}{8}$  inch-diameter PVC pipes on the x-axis. The total length of PVC pipes used during construction was 32.657 inches. The pipe was cut multiple times to incorporate 3 PVC cross tees that connect all the pipes linearly. The space between the first and second PVC cross tees, which represents the space between the reflector and driven element, is 7.940 inches. The space between the second and third PVC cross tees, which represents the space between the driven element and the director, is 12.441 inches. The center of each measuring tape wire was then connected to the PVC cross tees using two stainless steel hose clamps for each wire.

The driven element, however, had a slightly different construction process towards the end. Because this is where the voltage source will go, the two measuring tapes were individually connected to each side of the second PVC cross tee. Sandpaper was used to sand the plastic and paint off each tape, revealing their metal. Then, a wire was bent into a small U-shape and soldered onto both driven element halves. After the center conductor and coaxial cable were attached, the construction of the Yagi antenna was complete.



| Parts List |          |                                                     |
|------------|----------|-----------------------------------------------------|
| Item       | Quantity | Description                                         |
| 1          | 2        | 1/2" Sched 40 PVC Pipe-5in                          |
| 2          | 1        | 1/2" Sched 40 PVC Pipe-11in                         |
| 3          | 1        | 1/2" Sched 40 PVC Pipe-7in                          |
| 4          | 3        | 4-Way-PVC-Tee                                       |
| 5          | 2        | Cap-PVC-End                                         |
| 6          | 1        | 41.375(41-3/8)" Tape Measure                        |
| 7          | 2        | 17.75(17-3/5)" Tape Measure                         |
| 8          | 1        | 35.125(35-1/8)" Tape Measure                        |
| 9          | 1        | Hairpin wire 5" of #12 or #14 AWG Solid Copper Wire |
| 10         | 1        | X" RG-58A 50Ohm Coaxial Cable                       |
| 11         | 6        | 7/8" - 1-1/4" Stainless Hose Clamp                  |

Figure 1: Yagi Antenna Schematic

## Optimizations and Simulations

Before any construction could occur, the design had to first be simulated to ensure it worked. Using EZNEC, a digital schematic of our Yagi antenna was made. Figure 2 shows what the antenna looks like in a 3D digital environment. The reflector is the wire on the left, the driven element is the wire in the middle, and the director is the wire on the right. The driven element shows the voltage source V1 attached, as well as the L-network used to lower the VSWR.

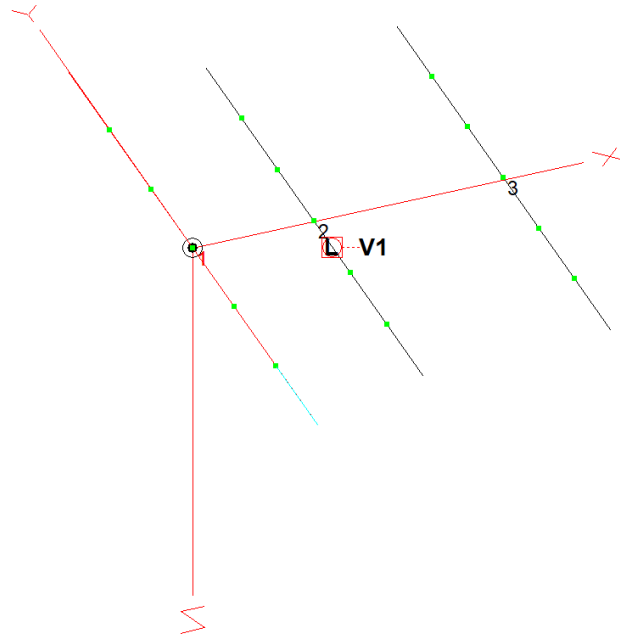


Figure 2: Yagi Antenna Schematic

Using this antenna, an azimuth plot or “far-field” plot was generated. Figure 3 shows the simulated azimuth plot, complete with frontal, back, and side lobes. This simulation reveals that the radiation is strongest in the frontal lobe, or the front direction of the antenna. This would mean that if the antenna were to be constructed, most of the radiated signal would be concentrated through this front side.

Total Field

EZNEC Demo

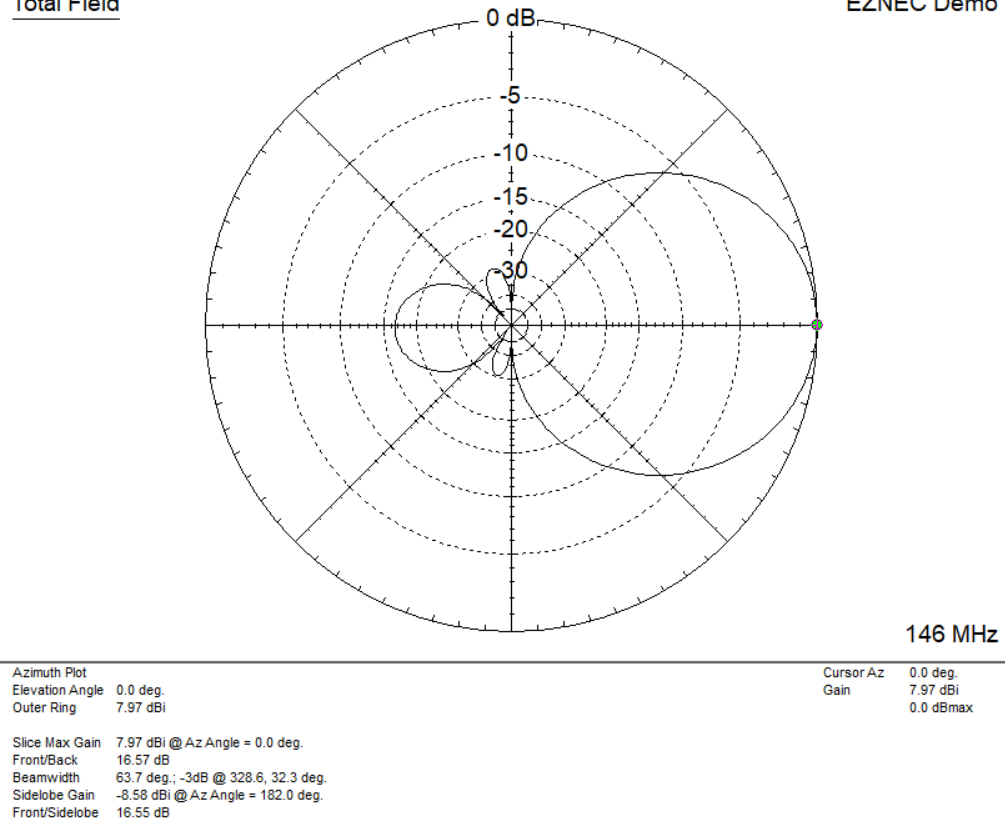


Figure 3: Yagi Antenna Azimuth Plot

When designing this antenna, it was clear that the VSWR was extremely high. This was because the antenna was not originally matched. However, using the “Circuit Design with Smith Chart” software, we were able to lower the VSWR value by adding a parallel inductor. Figure 4 shows the Smith chart used for matching. Then, the inductor was added in EZNEC under “L Networks” by adding the shunt branch (parallel branch). Finally, the length of the driven element was increased by a miniscule amount so that the VSWR could be lowered at 146 MHz. The simulated VSWR can be found in Figure 5. Here, a graph of the VSWR is shown between the frequencies of 144 MHz and 148 MHz. This graph shows us that the VSWR is lowest when the frequency is 146 MHz, with a VSWR value of 1.078.

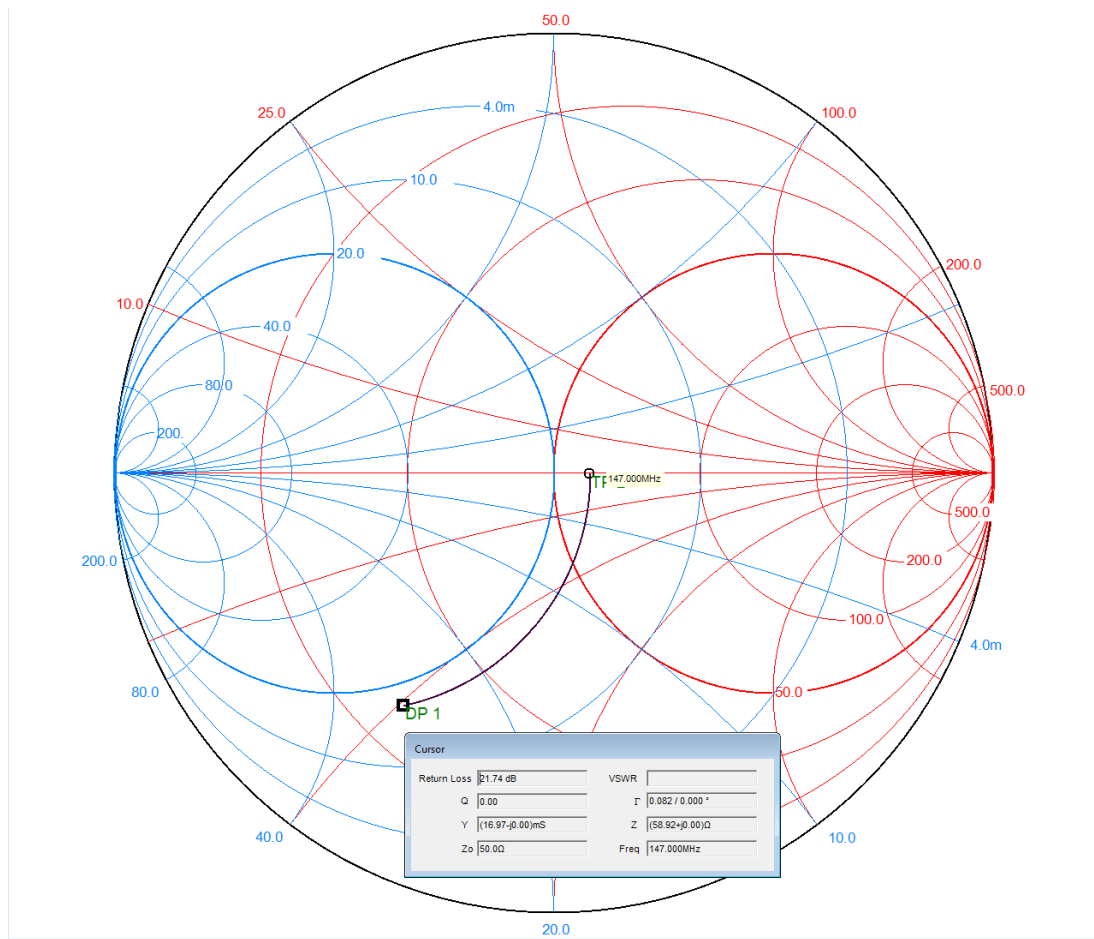


Figure 4: Smith Chart Matching with Parallel Inductor

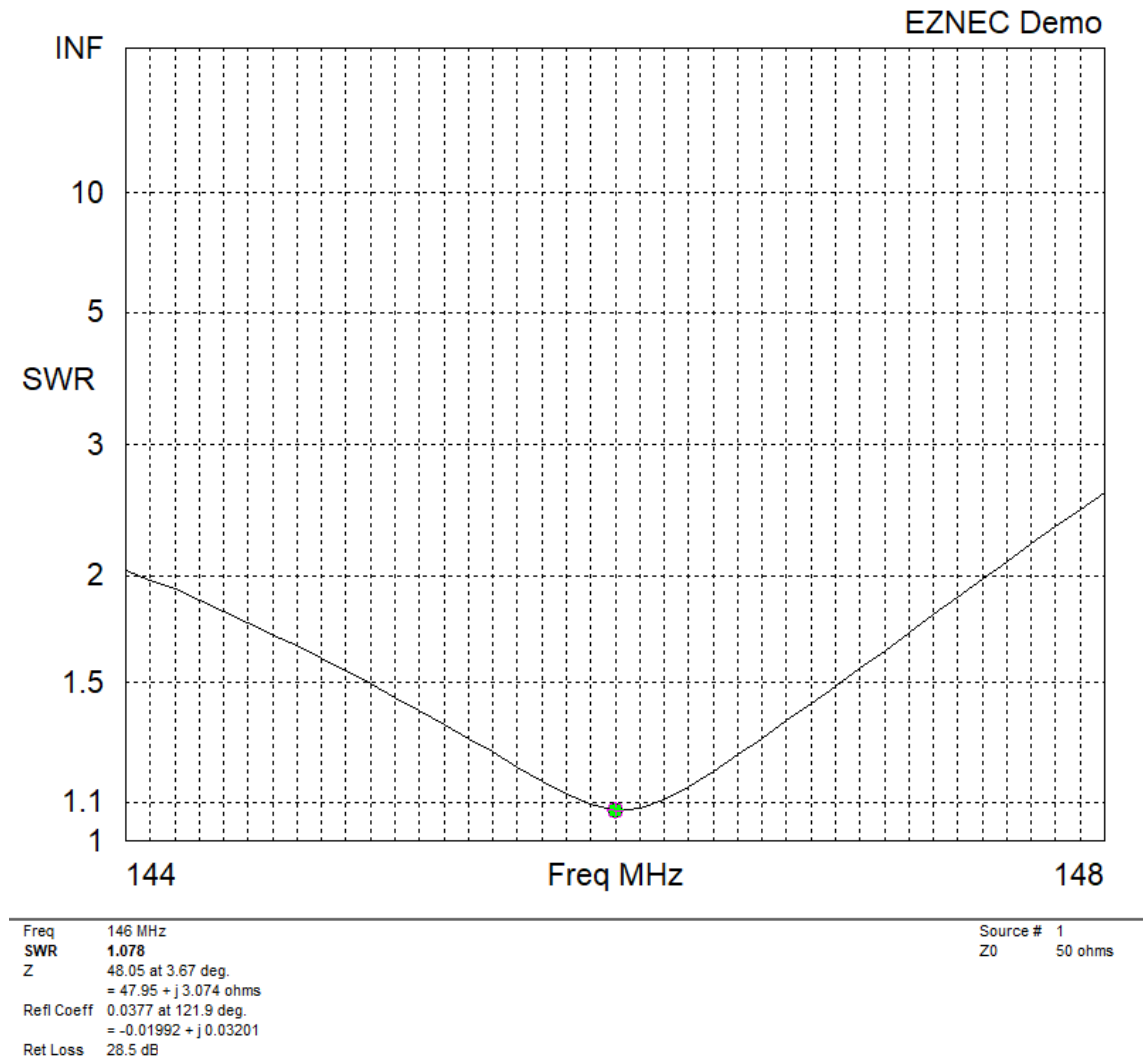


Figure 5: Smith Chart Matching with Parallel Inductor

## Testing and Results

### *Building a Half-Wave Dipole Antenna*

The calculated half-wavelength dipole is equal to  $\frac{\lambda}{2} = \frac{3 \times 10^8}{2 \times 440 \times 10^6} m = 34.091 \text{ cm}$ . We cut two wires with length  $\frac{\lambda}{4} = \frac{34.09 \text{ cm}}{2} = 17.045 \text{ cm}$ . These wires were connected to the screw terminals of the BNC to screw terminal adapter. This was our half-wave dipole antenna.

Before using the NanoVNA for measurements, we must calibrate the NanoVNA by connecting a short, open, and load to the coax cable connected to the NanoVNA. During this process, we must set the center frequency to 440 MHz and the span to 100 MHz. Note that the coax cable is wrapped around the balun to prevent current from flowing on the outside of the shield of the coax cable thereby preventing the coax cable from acting as an antenna of its own.

After calibrating our NanoVNA, we connected the NanoVNA to the half-wave dipole antenna to measure its VSWR and input impedance. Figure 6 displays the impedance

measurement of the half-wave dipole antenna. As displayed in Figure 6, the measured impedance of the half-wave dipole antenna is  $35.32 - j0.409$  ohms. Figure 7 displays the VSWR measurement of the half-wave dipole antenna. As displayed in Figure 7, the measured VSWR of the half-wave dipole antenna is 1.342. Figures 6 and 7 display that the resonant frequency of the half-wave dipole antenna is at 429 MHz.



Figure 6: Impedance measurement of half-wave dipole antenna



Figure 7: The VSWR measurement of half-wave dipole antenna

When placing our antenna close to a conductive surface, the input impedance value changed, the Smith chart looked distorted, and the VSWR measurements looked distorted. The input impedance value is unreliable when the antenna is placed near a conductive surface.

After cutting 1 cm from both ends of our antenna, the frequency with the minimum VSWR shifts toward the right. In other words, the resonant frequency becomes larger. The same behavior is depicted on the Smith chart where the frequency with the real input impedance value becomes larger.



### ***Building our own 146 MHz Yagi Antenna***

Before using the NanoVNA for measurements, we must calibrate the NanoVNA by connecting a short, open, and load to the coax cable connected to the NanoVNA. During this process, we must set the center frequency to 146 MHz and the span to 20 MHz. Note that the coax cable is wrapped around the balun to prevent current from flowing on the outside of the shield of the coax cable thereby preventing the coax cable from acting as an antenna.

After calibrating our NanoVNA, we connected the NanoVNA to our antenna to measure our antenna's VSWR and input impedance. This was measured at our antenna's operating frequency which was supposed to be 146 MHz. However, our antenna operated at 137.6 MHz (as displayed in Figures 11 and 12). Figure 11 shows the VSWR measurement of our antenna, which was about 1.997. Figure 12 displays the impedance of our antenna, which was  $39.05 + 35.4j$  ohms.



Figure 11: SWR measurement for our Yagi Antenna



Figure 12: Impedance for our Yagi Antenna

We also used the SDRuno Roadmap software to verify if our antenna was able to receive signals captured from the walkie-talkie. After verifying that our antenna received signals captured from the walkie-talkie, we used the software to measure the directive gain after positioning our antenna at different angles. Table 1 displays the directive gain,  $U(\phi)$ , measurements at different angles of  $\phi$ .

Table 1: Directive gain measurements at different angles

| Angle (in degrees) | Directivity Measurements (in dBm) |
|--------------------|-----------------------------------|
| 0                  | -40.2                             |
| 45                 | -37.1                             |
| 90                 | -54.5                             |
| 135                | -61.4                             |
| 180                | -53.7                             |
| 225                | -43.2                             |
| 270                | -88.9                             |
| 315                | -88.7                             |
| 360                | -85                               |

## Analyzed Results and Expected Values

### *Building our own 146 MHz Yagi Antenna*

Figure 11 displays the VSWR measurements of our antenna at its measured operating frequency (137.6 MHz). Figure 5 shows the expected VSWR measurements of our antenna at its expected operating frequency (146 MHz). The VSWR measurement displayed in Figure 11 is 1.997, which is slightly higher than the expected VSWR measurements (1.078) displayed in Figure 5. The VSWR is lower in Figure 5 because our simulated antenna was matched using an inductor, unlike our constructed antenna which was not matched using an inductor.

Figure 12 displays the measured impedance values of our antenna. The measured impedance value is equal to  $39.05 + 35.4j$  ohms which is more inductive and less resistive compared to the expected value (as displayed in Figure 5) of  $47.95 + 3.074j$  ohms. The measured antenna impedance is different from the expected antenna impedance most likely due to the placement of the antenna near field-interfering objects.

Table 1 displays the directive gain measurements at different angles. Figure 3 displays our yagi antenna's azimuthal plot. Figures 13 and 14 display the radar plots of the measured directive gain at different angles. Figures 13 and 14 were created by plotting the data in Table 1 using the radar plot chart in Google Sheets. In comparison to the results shown in Figure 3, the results shown in the radar plot look somewhat different. Figure 13 somewhat resembles the results displayed in Figure 3.

Despite the measurements not matching the expected values, our antenna did successfully fulfill its main purpose. Our antenna does successfully receive signals captured by the walkie-talkie. This was portrayed in the demo.

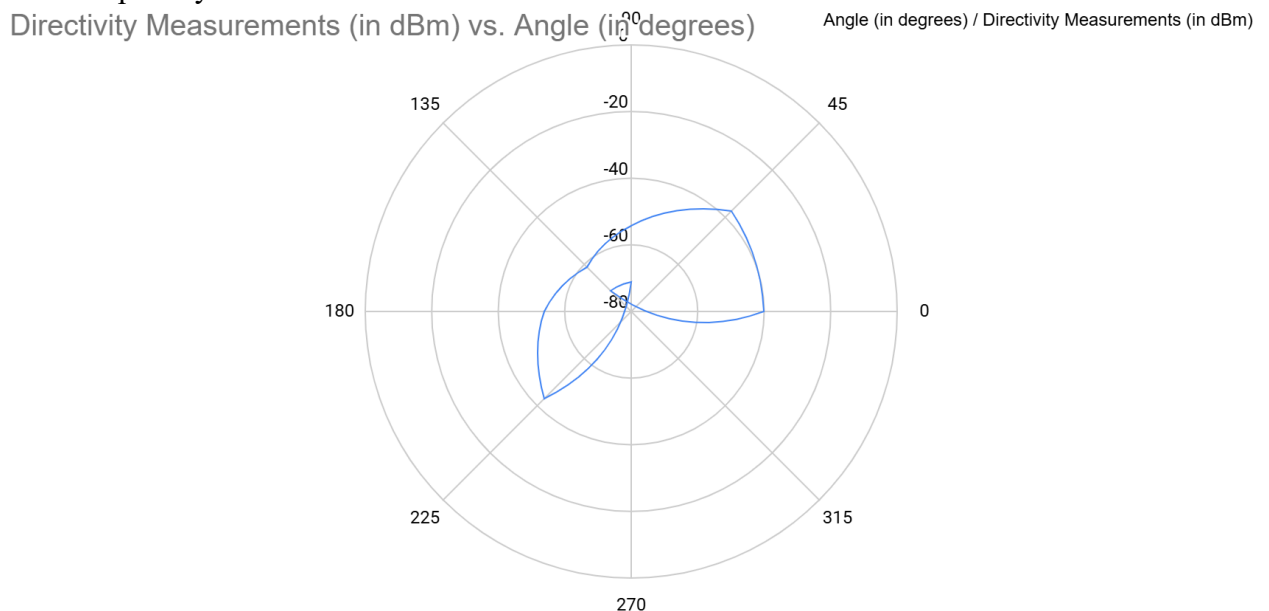


Figure 13: The Radar Plot for our measured directive gain at different angles without 360 degrees

Directivity Measurements (in dBm) vs. Angle (in degrees) / Angle (in degrees) / Directivity Measurements (in dBm)

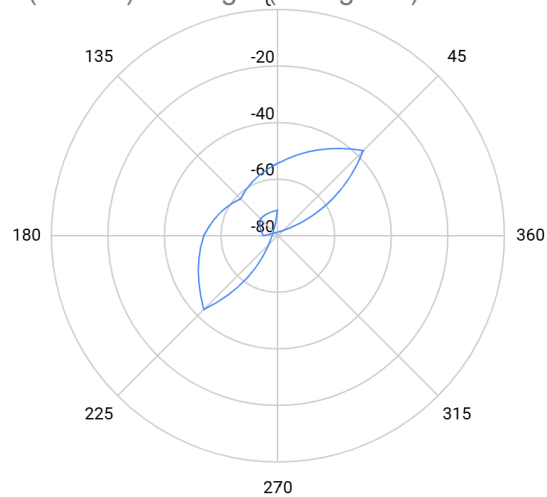


Figure 14: The Radar Plot for our measured directive gain at different angles without 0 degrees

## Conclusion

Two antennas were built during this lab; the half-wave dipole antenna and the yagi antenna. The half-wave dipole antenna was relatively simple to construct, while the yagi antenna took slightly more time to design and build. The VSWR was measured for each built antenna to determine how well the antenna matched with the feed line. Although the Yagi antenna's VSWR was slightly higher than our expected value, it was still low enough to work efficiently. Overall, we learned a lot about how antennas function, what some of their uses can be, how to build them, and the different types of antenna designs out there.

## Questions

### 1. Research baluns and explain why they are needed.

Baluns stand for “balanced to unbalanced”, and are used to separate balanced and unbalanced systems, such as the balanced antenna and the unbalanced coaxial cable feeder. This is to prevent the coaxial cable feeder from acting as its own antenna.

### 2. Look up the nominal input impedance of a half-wavelength dipole. Does this agree with what you measured? If not, why?

The nominal input impedance is  $73 + j42.5$  ohms. This does not agree with what we measured, as factors such as the diameter and proximity to ground affect the input impedance.

### 3. Why does shortening an antenna change its reactance?

Increasing the electrical length of the antenna increases its reactance. Therefore, decreasing the electrical length will decrease its reactance. This is because the physical length of the wire determines the resonance to the signal's wavelength, and if it is not in resonance, it will begin to act inductive or capacitive and thus have reactance.

**4. Why is an antenna affected by proximity to a conducting surface?**

Conducting surfaces will have electromagnetic fields that interact with the antenna, causing the radiation pattern and impedance to change.

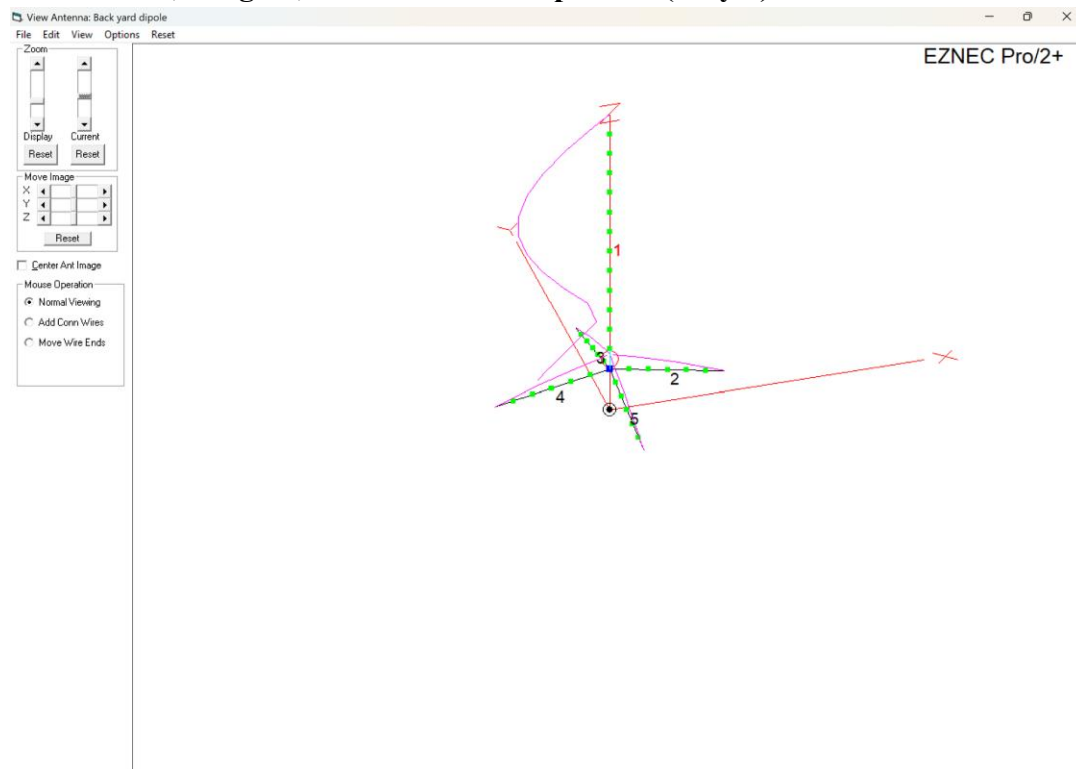
**5. Look up the type of antennas used in modern cell phones. Why are dipole antennas not used?**

Modern cell phones use inverted-F antennas because they can support several communication bands while still fitting within the tiny phone structure. Dipole antennas are not used because of their size and can not support several communication bans.

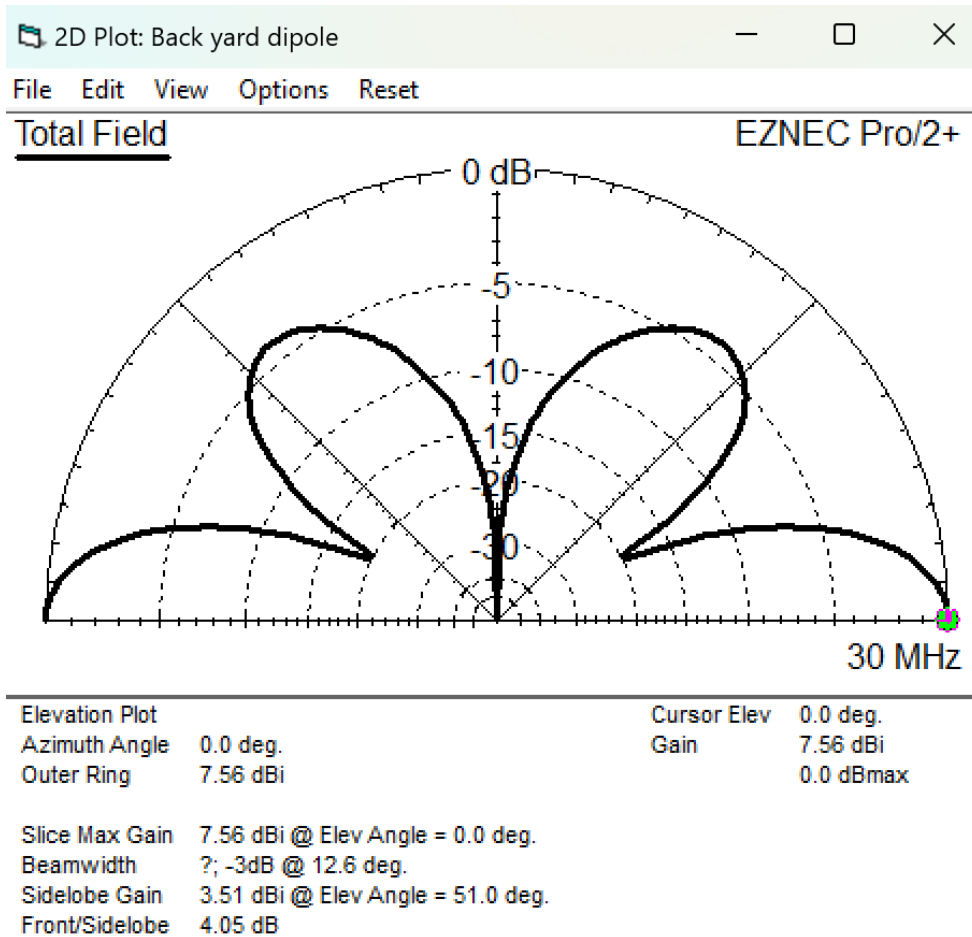
**6. What is a monopole antenna and how does it compare to a dipole?**

A monopole antenna is an antenna with one conductive element with a ground plane while a dipole element has two equal elements that feed into the center. It is typically half the size of a dipole antenna and is used in mobile phones and cars.

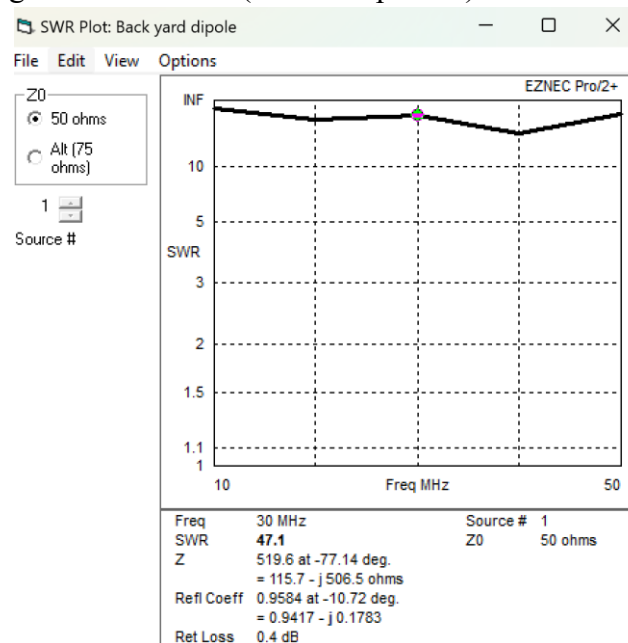
**7. Use the NEC software to analyze a  $5/8\lambda$  antenna above a ground plane. Assume a perfect ground. Also, assume that the feed point is at the bottom of the antenna between the antenna and ground. Find the input impedance, the current distribution, the gain, and the elevation pattern (vary  $\theta$ ) of this antenna.**



Above is an image of the current distribution on the  $5/8\lambda$  antenna



Above is an image of the azimuth (elevation pattern) for the  $5/8 \lambda$  antenna



Above is an image of the SWR and impedance measurement at different frequencies The maximum gain is 7.56 dBi and the input impedance is  $115.7 - j506.5$  ohms.

## References

- [1] Pasternack, "Yagi antennas," Pasternack, <https://www.pasternack.com/yagi-antennas-category.aspx> (accessed Dec. 11, 2024).
- [2] W. Wikipedia®, "Yagi–Uda Antenna," Wikipedia, [https://en.wikipedia.org/wiki/Yagi%E2%80%93Uda\\_antenna](https://en.wikipedia.org/wiki/Yagi%E2%80%93Uda_antenna) (accessed Dec. 11, 2024).
- [3] P. Bevelacqua, "Yagi-Uda Antenna," The Yagi-Uda Antenna - Yagi Antennas, <https://www.antenna-theory.com/antennas/travelling/yagi.php> (accessed Dec. 11, 2024).